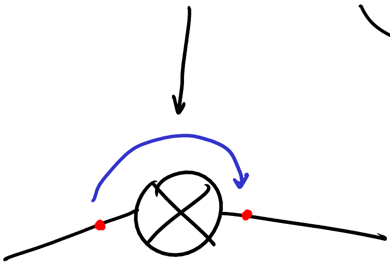
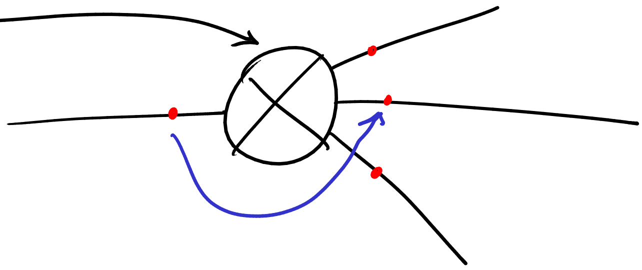
Lecture 9

Network Layer:

1. Sends from Sender to Receiver Hosts
2. Encapsulates into Packets
3. Does Forwarding & Routing





Forwarding is sending from input to output port of a router

choosing which port to forward to is done by Routing.

- Routing creates a Routing table which is used when Forwarding
- Any port of a Router can be both input and output port

Types of Network:

1. Virtual Circuit Network (VC Network) → Connection
2. Datagram Network → Connectionless

Datagram

- NO fixed end to end path
- No path Setup thus less overhead
- Only Destination Host address fixed

VC Network

- No longer used on the Internet
- Used in ATM, Frame relay, X-25

Routers:

- Ports can be both input and output
- Forwarding done by Switching Fabric
- Queuing: needed if more packets arrive than leave

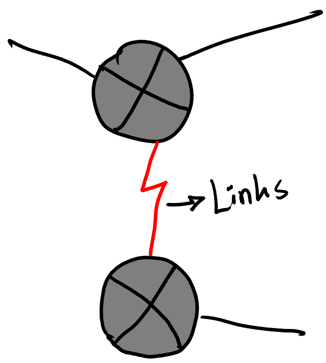
IP Protocol:

Data Length = Total Length - Header Length

TTL = Time To Leave, is a count of how many hops a data survives, substracted by 1 each time, used to avoid Routing loop when the target destination cannot be found

IP Fragmentation:

Network Links have MTU (Max Transfer Unit) which is why large packets are Fragmented before sending. Fragment order is identified by their respective Header bits



Packet / Datagram / Data

Packet / Datagram → Data + Header

Data → Data

- Total length of a Datagram = 5140, Header Length = 20, MTU = 1500, find the fragments.

Since MTU = 1500 & Total Data = 5140 - 20 = 5120

largest fragment of data = 1500 - 20 = 1980

	Data Bytes	Fragment Offset	DF	MF
∴ 1st sequence = 1480 + 20	0-1479	0/8 = 0	1	1
∴ 2nd sequence = 1480 + 20	1480-2959	1480/8 = 185	1	1
∴ 3rd sequence = 1480 + 20	2960-4439	2960/8 = 370	1	1
∴ 4th sequence = 640 + 20	4440-5119	4440/8 = 555	1	0

DF → Dont Fragment → if value = 1, router wont fragment data, expects fragmentation from the source

MF → More Fragments → if value = 1, means there are more fragments after this one left to be sent

ICMP

→ Internet Control Message Protocol
→ Helping Protocol for IP

- creates an error report for IP protocol
- Types:
  1. Ping
  2. TRACEROUTE

Ping:

- Used to teat the reachability of a host
- Sends 4 ICMP message request packet & waits for response
- Cam measure packet loss based on percentage of response

ICMP message:

Type	Code		
0	0	-	echo reply (ping)
8	0	-	echo request (ping)
3	3	-	port unreachable
11	0	-	TTL expired

ICMP can be used to figure out MTU by sending decreasing size of packets using Dont Fragment, as soon as no error is received, we have successfully sent the message without being dropped.

Ping Attacks:

1. DOS Attack / Ping Flood Attack
2. Zombie Attacks
3. DDOS Attack / Packet Magnification /(ICMP Smurf)

1. DOS:

- Denial of Service attack
- If the server is flooded with ping it crashes with hand excessive ping.
- Attacker varies source IP and sends numerous ping to the larget destination.

2. Zombie Attack:

- Attacker uses trojans or viruses to infect multiple PCs on the internet.
- The attacked PCs unaware, keep sending loads of ping to the target destination.
- The infected PCs are called zombies.

3. DDOS:

- Attacker sends ICMP echo packets to all the broadcast address of the targets nerwork with source set to the targets address.
- All the network responds to the target

TRACEROUTE:

An IMCP tool to trace path from source to destination host. It returns the IP Adress & Domain name.

working mechanism:

- The sourse sends echo requests with increasing TTL starting from 1
- Until the echo request reaches the target host, only TTL expired message (11, 0) is received
- When the host is reached, the unreachable port number prompts the message (3, 3) for Port Unreachable

Lecture 10 - DHCP, NAT, PAT

DHCP - Dynamic Host Configuration Protocol

Network Admins assign static IP address to some devices (Routers, Printers etc whose physical addresses arent likely to change, but others are assigned with DHCP.

- Routers, DHCP servers can provide Dynamic IP to devices
- They lease IPv4 addresses from their pool for a limited period of time(anywhere between a few hours to a week)
- The client must periodically contact the server to extend the lease.
- The server also checks if the client is present.

Steps to obtain a Lease (DORA)

1. Discover
2. Offer
3. Request
4. Acknowledgement

The new device Broadcasts a DHCP DISCOVER message with DHCP has port 67, so when broadcast message reaches any other device, they drop it, but DHCP server responds with Offer
Discover message: source: 0.0.0.0, dest: 255.255.255.255

The DHCP OFFER contains the IP address offered by the server by Unicast to the requesting device

The new device then Broadcasts a DHCP REQUEST message informing that it has accepted an IP, this lets the providing DHCP server confirm and lets any/all other DHCP providers know that their offer was not accepted, so they put the IP back in their pool.

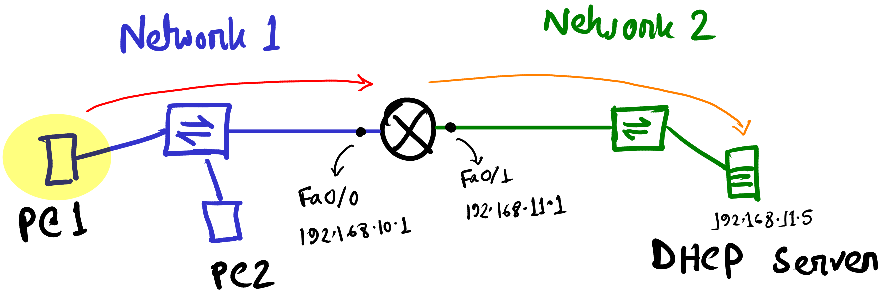
The DHCP server sends a Uncast message DHCPACK back to the new device confirming the Lease

Step 3 & 4 are done again when the client approaches Lease time. If the client gets a DHCPACK(4) after sending out a DHCPREQUEST(3) then the lease is renewed, otherwise(when some other device has taken the same address) all the 4 steps need to be done again for a new lease

Address:	Source	Destination	Type
Discover:	0.0.0.0	255.255.255.255	Broadcast
Offer:	DHCP Server IP	lease IP	Unicast
Request:	lease IP	255.255.255.255	Broadcast
Acknowledgement:	DHCP Server IP	lease IP	Unicast

DHCP Relay:

Discover messages are Broadcast messages, but when DHCP servers are in another network, the router will drop the message, because routers dont broadcast to other network without the target network's address, that's why DHCP Relay needs to be set in the router's default gateway



From PC1:

Dest MAC	Source MAC	Dest IP	Source IP	Dest Port	Source Port	DATA
FFFFFFFFFFFF	PC1	255.255.255.255	0.0.0.0	67	68	DATA

Relay from R1:

Dest MAC	Source MAC	Dest IP	Source IP	Dest Port	Source Port	DATA
DHCP Server	R1 Fa0/1 Mac	192.168.11.5	192.168.10.1	67	68	DATA

NAT - Network Address Translation

A way to translate Private addresses to Public addresses

Inside Network - the network of the sender host (as reference)
Outside Network - the network outside of the Border Gateway Router (from the host)

Inside Address - address of the host
Outside Address - address of the destination
Local Address - Private Address
Global Address - Public Address

∴ Inside Local Address : Private address of sender
∴ Inside Global Address: Public address of sender
∴ Outside Local Address : Private address of receiver
∴ Outside Global Address: Public address of receiver

Routers create NAT table when converting Inside address to Outside address, and delete the NAT table when the response is received

Advantages:
1. Helps save official IP addresses
2. Keeps internal network settings consistent, even when public IP configurations change.
3. Keeps user and device IPs private from the outside internet, enhancing security.

Disadvantages:
1. Increases Delay
2. End-to-End Traceability is lost
3. Can cause problems for some protocols

PAT - Port Address Translation

- Allows many devices (more than NAT) to share the same IP by using Port Addresses
  ex: Inside Local Address: 192.168.10.11:1331 → Inside Global Address: 209.165.200.226:1331

- Uses port number to differentiate the devices.

- Router uses a PAT Table: Private IP + Port ⟶ Public IP + new Port

- PAT changes both IP Address and Port number

Disadvantages:

- Since some protocols like ICMP doesnt have access to Port Numbers, PAT uses Query ID instead

NAT Table with Overload

Inside Local IP Address	Inside Global IP Address
192.168.10.11:1444	209.165.200.226:1444
192.168.10.12:1444	209.165.200.226:1445

in case of devices having same port addresss
the NAT table uses the next available port address

Port Forwarding:

- Needed when someone tries to access a Private Address from the Internet

- Router is configured to route Public Addresses to Private ones

- Router maps PublicIP:Port to a corresponding PrivateIP:Port